

Ideation Phase

Brainstorm & Idea Prioritization Template

Date	09 March 2026
Team ID	
Project Name	Electric Vehicle Data Analytics and Visualization using Tableau
Maximum Marks	4 Marks

Step-1: Team Gathering, Collaboration and Select the Problem Statement

Problem Statement

With the increasing adoption of Electric Vehicles (EVs), there is a growing need to analyse charging patterns, battery performance, and vehicle efficiency. However, large volumes of EV data are difficult to interpret without proper visualization tools. This project aims to develop interactive dashboards using Tableau to analyse EV charging behaviour, battery health, and model efficiency to support decision-making for city planners, fleet managers, and EV consumers.

Team Collaboration

Our team discussed various challenges related to electric vehicles such as charging infrastructure demand, battery health monitoring, and EV model comparison. After brainstorming different ideas, we selected EV data analytics and visualization because it provides useful insights for multiple stakeholders and helps make data-driven decisions.

Step-2: Brainstorm, Idea Listing and Grouping

Brainstormed Ideas

1. Analyze EV charging patterns by time and location.
2. Monitor battery performance and charging cycles.
3. Compare predicted vs actual driving range.
4. Identify peak EV charging hours in urban areas.
5. Compare efficiency of different EV models.
6. Calculate cost per kilometer for EV vehicles.
7. Analyze environmental impact and emission reduction.

Idea Grouping

Charging Infrastructure Analysis

- Charging time patterns
- Peak charging hours
- Charging station demand
- Battery and Vehicle Performance

- **Battery health monitoring**
- **Driving range analysis**
- **Charging cycles and efficiency**
- **Consumer Decision Support**
- **EV model comparison**
- **Cost per kilometer**
- **Environmental benefits**

Step-3: Idea Prioritization

Selected Ideas

1. Charging Pattern and Usage Analysis
2. Battery Performance vs Driving Range Dashboard
3. Comparative EV Model Efficiency Dashboard

Reason for Selection

These ideas were selected because they provide valuable insights for city planners, fleet managers, and EV buyers. They help analyze charging demand, monitor battery performance, and compare EV models using interactive Tableau dashboards.